

Institute of Actuaries of India

Subject CM1A – Actuarial Mathematics (Paper A)

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: Option B**[4]***(No marks for working)*The present value of payment as at 01/01/2023 is $5000 \ddot{a}_{48|}^{(12)}$

$$\begin{aligned} \text{So, the present value as at 01/01/2022 is } 5000v \ddot{a}_{48|}^{(12)} &= 5000 * v * \frac{1-v^{(12).48}}{d^{(12)}} \\ &= 5000 * 0.917431 * \frac{0.29157}{.0072} = \text{Rs. } 186,912 \end{aligned}$$

Solution 2:

- i) Discounted payback period (DPP): The discounted payback period for a project is the smallest time t at which the present value or the accumulated value of the income up to time t exceeds the present value or the accumulated value of the outgoes up to time t . (1.5)
- Payback period: Earliest time at which the total value of the income received to date is greater than or equal to the total value of outgo to date. Since interest is ignored, it takes no account of time value of money. (1.5)
- ii) Disadvantages are: [3]
- Neither the DPP nor the PP gives any indication of how profitable the project is, as they do not take into account the cashflows happening once the DPP or PP is achieved (1)
 - PP can be misleading as it does not consider time value of money (1)
 - There may not be unique DPP or PP (1)
- iii) When solving for IRR, there may not be a unique solution or there may be multiple solutions. In these cases, the result is hard to interpret. (1)
- Does not indicate the amount of profits the project will generate (1)
- [7 Marks]

Solution 3: Option A**[3]***(No marks for working)*

The expected present value is

$$\begin{aligned} 50000 10|A_{50} &= 50000 v^{10} 10p_{50} A_{60} \\ &= 50000 1.06^{-10} \times \frac{l_{60}}{l_{50}} \times A_{60} \\ &= 50000 * 1.06^{-10} \times \frac{9287.2164}{9712.0728} \times 0.32692 \\ &= \text{Rs. } 8728 \end{aligned}$$

Solution 4: Option C**[4]***(No marks for working)*The variance of the benefit from this annuity is: $Var[a_{k70|}] = \frac{500^2}{d^2} [2A_{70} - (A_{70})^2]$

Using values from the tables, the variance comes as:

$$\begin{aligned} &= \frac{500^2}{\left(\frac{.06}{1.06}\right)^2} [0.26583 - (0.47743)^2] = 2956519 \\ \text{Std. deviation} &= \sqrt{2956519} \\ &= 1719 \end{aligned}$$

Solution 5: Option C [3]

Solution 6: Option C [4]

Workings for reference

$$\begin{aligned} \text{EPV of no claim bonus is as below} &= 15,000 * e^{-\int 4060(0.03+0.04) dx} * e^{-\int 40600.05 dx} \\ &= 15,000 * e^{(-0.07*20)} * e^{(-0.05*20)} \\ &= 15,000 * 0.2466 * 0.3679 \\ &= 1,360.86 \end{aligned}$$

Solution 7: Health insurance contracts pay benefits contingent upon the state of health of a person. (1)

Critical illness plan pays lumpsum on diagnosis of a defined critical illness (1)

Long term care insurance pays income contingent upon policyholder requiring long term care, hence supports the costs of receiving that care. (1)

Income protection – pays income to the policyholder while that policyholder is deemed as being “sick”. The definition of “sick” is carefully specified in the policy conditions. (1)

If the policyholder recovers, the cover under the policy usually continues, so that subsequent bouts of qualifying sickness would merit further benefit payments (1)

Such policies are usually subject to a deferred period (say 3 months) of continuous sickness that has to have elapsed before any payments start to be paid, and during which no benefit is paid (1)

Premium are usually regular and no premium payment is required in periods of qualifying sickness (1)

[7 Marks]

Solution 8: In assessing the suitability, it is important to consider the following:

- i) Objective of the modelling exercise (0.5)
- Validity of the model for the purpose to which it is put (0.5)
- Validity of the data to be used (0.5)
- Validity of the assumption (0.5)
- Possible errors associated with the model or parameters not being a perfect representation of the real world situation being modelled (0.5)
- Impact of correlations between random variables that drive the model (0.5)
- Extent of correlations between various results produced from the model (0.5)
- Current relevance of the models written and used in the past (0.5)
- Credibility of data input (0.5)
- Credibility of results output (0.5)
- Dangers of spurious accuracy (0.5)
- Ease with which model and its results can be communicated (0.5)
- Regulatory requirements (0.5)

[Max 5]

- ii) Issues noted:
- Number of policies should reduce throughout (due to expected deaths), but these increase in year 4.
- Projection is provided for 8 years, while policy term is 10 year. So some cash flows are missing.
- Death benefit should be multiple of premium plus accrued bonus. But the death benefit appears to stay constant across the projection i.e. bonuses don't seem to be included.
- Premium term is mentioned as 5 years, but 7 premiums are output by the model.
- Commission outgo is expected in year 8, while there is no premium income in this year.
- Other cash flows such as death outgo and maturity outgo seem to truncate in year 7.
- Premium amount is 60,000, but it can only take values of 50,000 or 1,00,000

[7]

[12 Marks]

Solution 9: Capital gain test:

$$\left(1 + \frac{i^{(4)}}{4}\right)^4 = 1.04 \Rightarrow i^{(4)} = 0.039414 \quad (1)$$

$$(1 - t_1) \times \frac{D}{R} = (1 - 0.2) \times \frac{.05}{1.03} = .038835 \quad (1)$$

$$\Rightarrow i^{(4)} > (1 - t_1) \times \frac{D}{R} \quad (0.5)$$

$$\Rightarrow \text{Capital gain on the contract} \quad (0.5)$$

$\Rightarrow$ The worst-case result would be that bond is redeemed by borrower as late as possible (i.e: after 20 years). This will give the minimum yield; (0.5)

$$P = 1,00,000 \times 0.05 \times 0.80 \times a_{20|}^4 + (103000 - 0.25(103000 - P)) \times v^{20} \quad (1.5)$$

$$\Rightarrow P = \frac{4000a_{20|}^4 + 77250v^{20}}{1 - 0.25v^{20}} \quad (0.5)$$

$$a_{20|}^4 = a_{20|} \times \frac{i}{i^4} = 13.5903 \times 1.014877 = 13.79248 \quad (0.5)$$

$$\Rightarrow \frac{4000 \times 13.79248 + 77250 \times 0.45639}{1 - 0.25 \times 0.45639} = 1,02,072.25 \quad (1)$$

[7 Marks]

Solution 10:

i) The endowment assurance is calculated as:

$$\bar{A}_{25:25|} = \bar{A}_{25:25|} + A_{25:25|} \approx 1.06^{0.5} A_{25:25|} + v^{25} 25p_{25} \quad (1)$$

$$v^{25} 25p_{25} = v^{25} \frac{l_{50}}{l_{25}} = 1.06^{-25} \times \frac{9712.0728}{9953.6144} = 0.22734 \quad (0.5)$$

$$A_{25:25|} = A_{25} - v^{25} 25p_{25} A_{50} = .010386 \quad (0.5)$$

$$\text{So, } \bar{A}_{25:25|} \approx 1.06^{0.5} \times .010386 + 0.22734 = 0.23804 \quad (1)$$

The temporary annuity $\bar{a}_{25:25|}$ is calculated as:

$$\bar{a}_{25:25|} = \bar{a}_{25} - v^{25} 25p_{25} \bar{a}_{50} \approx (16.660 - 0.5) - 0.22734 \times (14.044 - 0.5) = 13.081 \quad (2)$$

[5]

ii) Following premium conversion relationship is expected to hold between the identities

$$\bar{A}_{25:25|} = 1 - \delta \bar{a}_{25:25|}$$

$$\bar{A}_{25:25|} = 1 - \ln 1.06 \times 13.081 = 0.23779$$

The answers are approximately equal. The slight discrepancy is due to different assumptions underlying the two calculations.

[2]

[7 Marks]

Solution 11:

- i)
$$L = 50000v + 48000v^2 + 46000v^3 + \dots + 12000v^{20} \quad (1)$$

$$= 52000(v + v^2 + v^3 + \dots + v^{20}) - 2000(v + 2v^2 + 3v^3 + \dots + 20v^{20})$$

$$= 52000 a_{\overline{20}|} - 2000(Ia)_{\overline{20}|} \quad (1)$$

$$52000 * 13.5903 - 2000 * 125.155 = \text{Rs. } 456385.60 \quad (1)$$

[3]
- ii) **Interest component in 12th instalment:**
- Amount of 12th instalment = $50,000 - 2000 * 12 = 28000$ (0.5)
 Loan outstanding after 11th instalment
- $$L_{11} = 28000v + 26000v^2 + \dots + 12000v^9 \quad (0.5)$$
- $$= 30000 a_{\overline{9}|} - 2000(Ia)_9 \quad (0.5)$$
- $$= 30000 * 7.4353 - 2000 * 35.2366 = \text{Rs. } 152585.80 \quad (0.5)$$
- Interest component in 12th instalment = $\text{Rs. } 152585.80 * .04 = \text{Rs. } 6103.43$ (0.5)
 Hence, Capital repaid in 12th instalment = $28000 - 6103.43 = \text{Rs. } 21896.57$ (0.5)
[3]
- iii) **Remaining term of Loan:**
- Loan o/s after 12th instalment = $152585.80 - 21896.57 = \text{Rs. } 130689.23$ (1)
 Level annual instalment = $\text{Rs. } 28000$ (0.5)
- Remaining term is given by equation:
- $$130689.23 = 28000 * a_{\overline{n}|} \quad (0.5)$$
- $$a_{\overline{n}|} \geq \frac{130689.23}{28000} \text{ i.e. } a_{\overline{n}|} \geq 4.6675 \quad (0.5)$$
- $$a_{\overline{5}|} = 4.4518 \text{ \& } a_{\overline{6}|} = 5.2421 \quad \text{Hence } n = 6 \quad (0.5)$$
-
- [3]**
- iv) **Calculating Final reduced payment**
- Let R be the final payment
- $$130689.23 = 28000 * a_{\overline{5}|} + Rv^6 \quad (1)$$
- $$130689.23 = 28000 * 4.4518 + 0.79031R \quad (1)$$
- $$R = \frac{130689.23 - 28000 * 4.4518}{0.79013} = \text{Rs. } 7641.09 \quad (1)$$
-
- [3]**
- v) **Interest paid during term of loan**
- Total amount paid by way of annual instalments = $\text{Rs. } 615641.09$ (1)
 Loan amount = $\text{Rs. } 456385.60$
- Interest paid = $615641.09 - 456385.60 = \text{Rs. } 159255.49$ (1)
[2]
[14 Marks]

Solution 12:

i) Future loss random variable (FLRV)

FLRV at time zero is:

$$\left\{ \begin{array}{ll} (3,00,800 - 15000K_{50})v^{T_{50}} + 500 + 50a_{\overline{K}_{50}|} - P \ddot{a}_{\overline{K}_{50}+1|} & K_{50} < 15 \\ 500 + 50a_{14} - P \ddot{a}_{15|} & K_{50} \geq 15 \end{array} \right. \quad (2)$$

Just before the payment of 10th premium, the life is aged 59 years, and the balance of term is 6 years.
So, the FLRV is now:

$$\left\{ \begin{array}{ll} (1,65,800 - 15000 K_{59})v^{T_{59}} + 50a_{\overline{K}_{59}+1|} - P \ddot{a}_{\overline{K}_{59}+1|} & K_{59} < 6 \\ 50 \ddot{a}_{6|} - P \ddot{a}_{6|} & K_{59} \geq 6 \end{array} \right. \quad (2)$$

[4]

ii) The premium equation using the principle of equivalence is:

$$P a_{50:\overline{15}|} = 315800 \bar{A}_{50:\overline{15}|} - 15000 (\bar{IA})_{50:\overline{15}|} + 500 + 50 (a_{50:\overline{15}|} - 1)$$

$$a_{50:\overline{15}|} = 11.253 \text{ (From the tables)}$$

$$\bar{A}_{50:\overline{15}|} \approx (1+i)^{0.5} A_{50:\overline{15}|} = 1.04^{0.5} [A_{50:\overline{15}|} - \frac{D_{65}}{D_{50}}]$$

$$\frac{D_{65}}{D_{50}} = \frac{689.23}{1366.61} = 0.50434$$

$$= 1.04^{0.5} [0.56719 - 0.50434] = 0.06410$$

The increasing term assurance factor is

$$(\bar{IA})_{50:\overline{15}|} \approx (1+i)^{0.5} (IA)_{50:\overline{15}|} = 1.04^{0.5} [(IA)_{50} - \frac{D_{65}}{D_{50}} ((IA)_{65} + 15A_{65})]$$

$$= 1.04^{0.5} \times [8.55929 - 0.50434 \times (7.89442 - 15 \times 0.52786)] = 0.59615$$

Hence Premium P is:

$$11.253 P = 315800 \times 0.06410 - 15000 \times 0.59615 + 500 + 50 \times (11.253 - 1)$$

$$= Rs. 1094.19$$

[6]

iii)

$${}_9V = 180800 \bar{A}_{59:\overline{6}|} - 15000 (\bar{IA})_{59:\overline{6}|} - (1094.19 - 50)a_{59:\overline{6}|} \quad (1)$$

$$\bar{A}_{59:\overline{6}|} \approx 1.04^{0.5} A_{59:\overline{6}|} = 1.04^{0.5} [A_{59:\overline{6}|} - \frac{D_{65}}{D_{59}}] \quad (0.5)$$

$$\begin{aligned} \frac{D_{65}}{D_{59}} &= \frac{689.23}{924.60} = 0.745307 \\ &= 1.04^{0.5} [0.79446 - 0.745307] = 0.05013 \end{aligned} \quad (0.5)$$

And increasing term assurance factor is:

$$(\bar{IA})_{59:\overline{6}|} \approx (1.04)^{0.5} (IA)_{59:\overline{6}|} = 1.04^{0.5} [(IA)_{59} - \frac{D_{65}}{D_{59}} ((IA)_{65} + 6A_{65})] \quad (0.5)$$

$$1.04^{0.5} [8.42588 - 0.745307 \times (7.89442 - 6 \times 0.52786)] = 0.18520 \quad (0.5)$$

$$\text{Annuity function } a_{59:\overline{6}|} = 5.344$$

The reserve is:

$$9V = 180800 \times 0.05013 - 15000 \times 0.18520 - 1044.19 \times 5.344 = 704.66 \quad (0.5)$$

Assumption: Reserving basis are the same as that used to calculate the premium (0.5)

[4]

[14 Marks]

Solution 13:

i) The profit vector is the expected profit at the end of each policy year per policy in force at the beginning of that policy year. (1)

The vector of expected profit per policy issued is called the profit signature, For example if (PRO)_t is the profit vector, ${}_t-1P_x \cdot (\text{PRO})_t$ is the profit signature, where ${}_t-1P_x$ is the probability of the policy issued to a life aged x is in force at the beginning of year t . (1)

[2]

ii)

Age (x)	Qx^d	Qx^s	$(aq)_x^d$	$(aq)_x^s$	(ap)	${}_t-1(ap)$
50	0.001971	10%	0.001971	0.09980	0.898226	1
51	0.002732	10%	0.002732	0.09973	0.897541	0.898226
52	0.003152	10%	0.003152	0.09968	0.897163	0.806195
53	0.003539	0%	0.003539	0.00000	0.996461	0.723288

(2)

Year	Profit Vector	Probability In-Force	Discount Factor	Expected Profit	Premium Signature
	From Ques	${}_t-1(ap)$	using RDR	$a*b*c$	$P*{}_t-1(ap)*\text{Disc}$
1	7,211.25	1	0.930233	6,708.14	15,000.000
2	(1,262.18)	0.8982261	0.865333	(981.05)	12,533.387
3	(1,063.56)	0.80619493	0.804961	(690.20)	10,464.401
4	(6,082.67)	0.72328842	0.748801	(3,294.37)	8,733.280
			Total	1,742.519	46,731.069

(3)

$$\text{Profit Margin} = 1,742.519 / 46,731.069 = 3.73\%$$

(1)

[6]

iii) Option B

(4)

iv)

Year	Zeroised Profit Vector	Probability in Force	Discount Factor	PV
1	1,553.31	1	0.93023	1,444.94
2	0	0.898226	0.86533	-
3	0	0.806195	0.80496	-
4	0	0.723288	0.74880	-
			Total PV	1,444.94

$$\text{Profit Margin} = 1,444.94 / 46,731.069 = 3.09\%$$

(2)

[14 Marks]
